QAL Quantum Inspire Hybrid Computing Workshop

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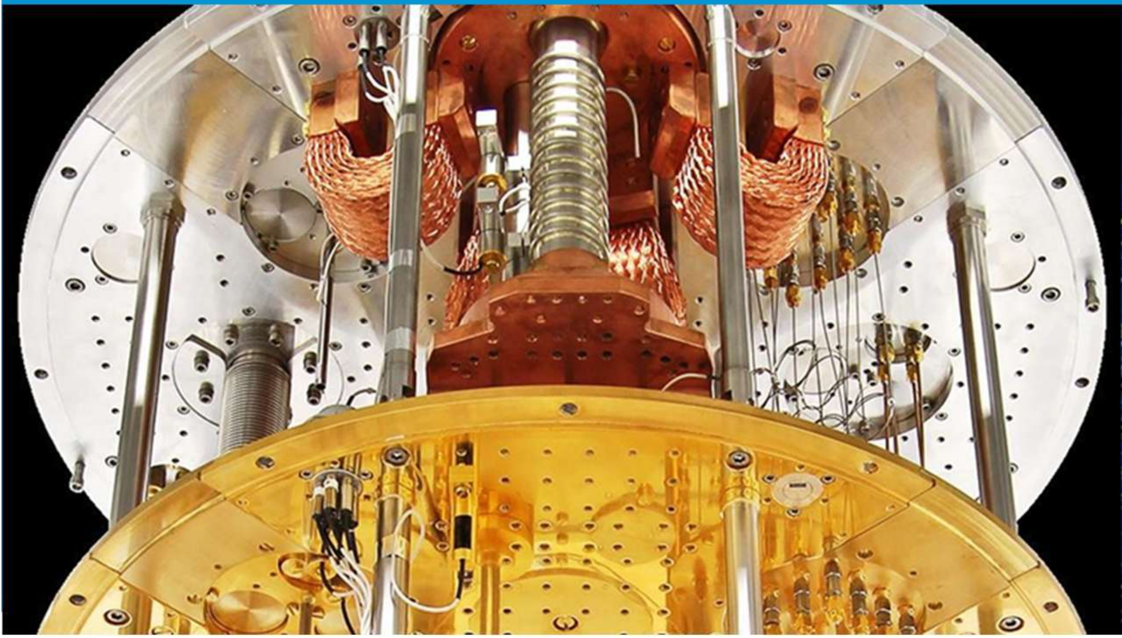


H2 Hybrid Algorithm

QuTech

QuTech is a collaboration between TU Delft TNO

Context

Quantum computing could realize earliest use cases in these 4 industries (McKinsey)

1. pharmaceuticals
2. chemicals
3. automotive
   including battery materials and hydrogen storage
4. finance

Simulation of molecules / materials plays a large role

Quantum Phase Estimation (QPE) requires many qubits and many operations (high circuit width and depth) ➔ not suitable for NISQ devices.

Variational (hybrid) algorithms require fewer qubits and operations, but more measurements and more shots ➔ e.g. Variational Quantum Eigensolver (VQE)

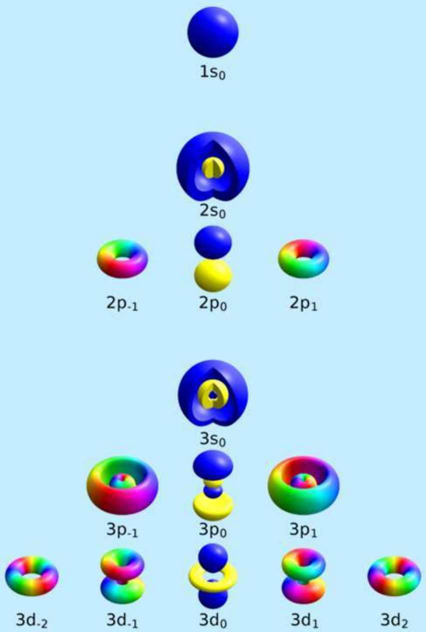
Problem

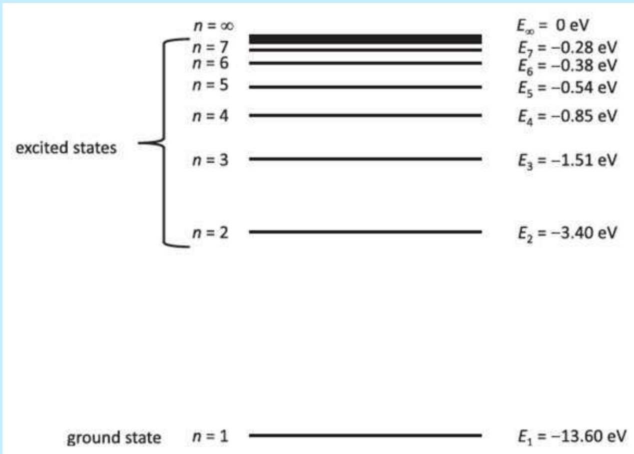
"What is the energy of the two electrons of the hydrogen molecule (in their ground state)?"

From atoms to molecule

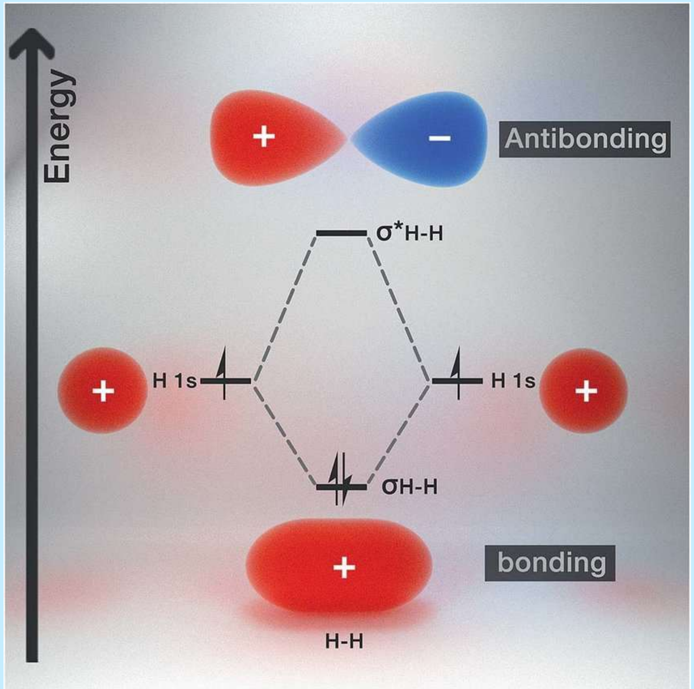
The electrons in the hydrogen atom are described by a wavefunction or orbital

The quantum numbers (n, l, m) determine the shape and energy of the allowed electronic states





When two atoms are brought together, the orbitals get mixed, yielding new shapes and energies



Calculating energies

The energy of the electrons is described by the Hamiltonian operator

$$\hat{H} = \hat{T}_e + \hat{V}_{ne} + \hat{V}_{ee},$$

$$\hat{T}_e = -\sum_i \frac{\hbar^2}{2M_i}\nabla_i^2,$$

$$\hat{V}_{ne} = -\sum_{i,k} \frac{e^2}{4\pi\epsilon_0}\frac{Z_k}{|\mathbf{r}_i - \mathbf{R}_k|},$$

$$\hat{V}_{ee} = \frac{1}{2}\sum_{i\neq j} \frac{e^2}{4\pi\epsilon_0}\frac{1}{|\mathbf{r}_i - \mathbf{r}_j|},$$

Using second quantization, we can describe the Hamiltonian with annihilation (a) and creation ($a^\dagger$) operators

$$\hat{H} = \sum_{pq} h_{pq}\hat{a}_p^\dagger \hat{a}_q + \frac{1}{2}\sum_{pqrs} h_{pqrs}\hat{a}_p^\dagger \hat{a}_q^\dagger \hat{a}_r \hat{a}_s,$$

From electrons to qubits

The Jordan-Wigner encoding maps the occupancy of each molecular orbital (empty/filled) to a qubit state ($|0\rangle/|1\rangle$)

$$\hat{a}_j^\dagger \rightarrow \frac{X_j - iY_j}{2} \otimes Z_0 \otimes \cdots \otimes Z_{j-1}$$
$$\hat{a}_j \rightarrow \frac{X_j + iY_j}{2} \otimes Z_0 \otimes \cdots \otimes Z_{j-1}.$$

The parity encoding maps the parity of the occupancy of lower-energy orbitals to a qubit state ($|0\rangle/|1\rangle$)

$$\hat{a}_j^\dagger \rightarrow \frac{Z_{j-1} \otimes X_j - iY_j}{2} \otimes X_{j+1} \otimes \cdots \otimes X_{n-1}$$
$$\hat{a}_j \rightarrow \frac{Z_{j-1} \otimes X_j + iY_j}{2} \otimes X_{j+1} \otimes \cdots \otimes X_{n-1}$$

There are other encodings

Measurement

The Hamiltonian can now be represented as a sum of Pauli strings $\hat{P}_a$

$$\hat{H} = \sum_a w_a \hat{P}_a$$

e.g. $\hat{P}_1 = X \otimes X \otimes Y \otimes X$

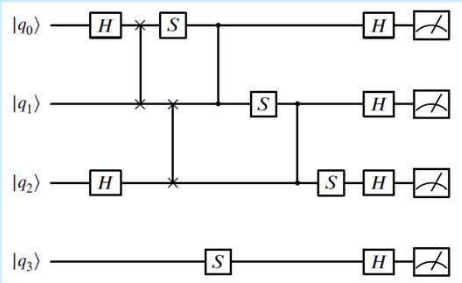
To measure in the Z-basis, X and Y need to be rotated

$$Z = \mathrm{Ry}\left(-\frac{\pi}{2}\right) X \mathrm{Ry}\left(-\frac{\pi}{2}\right)^\dagger$$

$$Z = \mathrm{Rx}\left(\frac{\pi}{2}\right) Y \mathrm{Rx}\left(\frac{\pi}{2}\right)^\dagger$$

To measure multiple Pauli strings simultaneously, a unitary operator U can be applied beforehand

$$U^\dagger (XXYX) U = ZIII$$
$$U^\dagger (IIYX) U = IZII$$
$$U^\dagger (YYYX) U = IIZI$$
$$U^\dagger (IIXY) U = IIIZ$$



Trial states (Ansätze)

Finding the ground-state energy =
finding the state that gives the lowest
eigenvalues of the Hamiltonian operator

$$E_0 = \min_{\theta} \left\langle \psi(\theta) \middle| \hat{H} \middle| \psi(\theta) \right\rangle$$

The trial state (or Ansatz) $|\psi(\theta)\rangle$

- is determined by parameters θ
- can be found by applying a unitary operation $U(\theta)$ to the initial state $|0\rangle$

$$|\psi(\theta)\rangle = U(\theta)|0\rangle$$

Popular Ansätze are variations on

- the Hardware-Efficient Ansatz (HEA)
- the Unitary Coupled-Cluster (UCC) Ansatz

Optimization

Finding the optimal parameters θ is done with classical optimization

Different types of optimizers are:

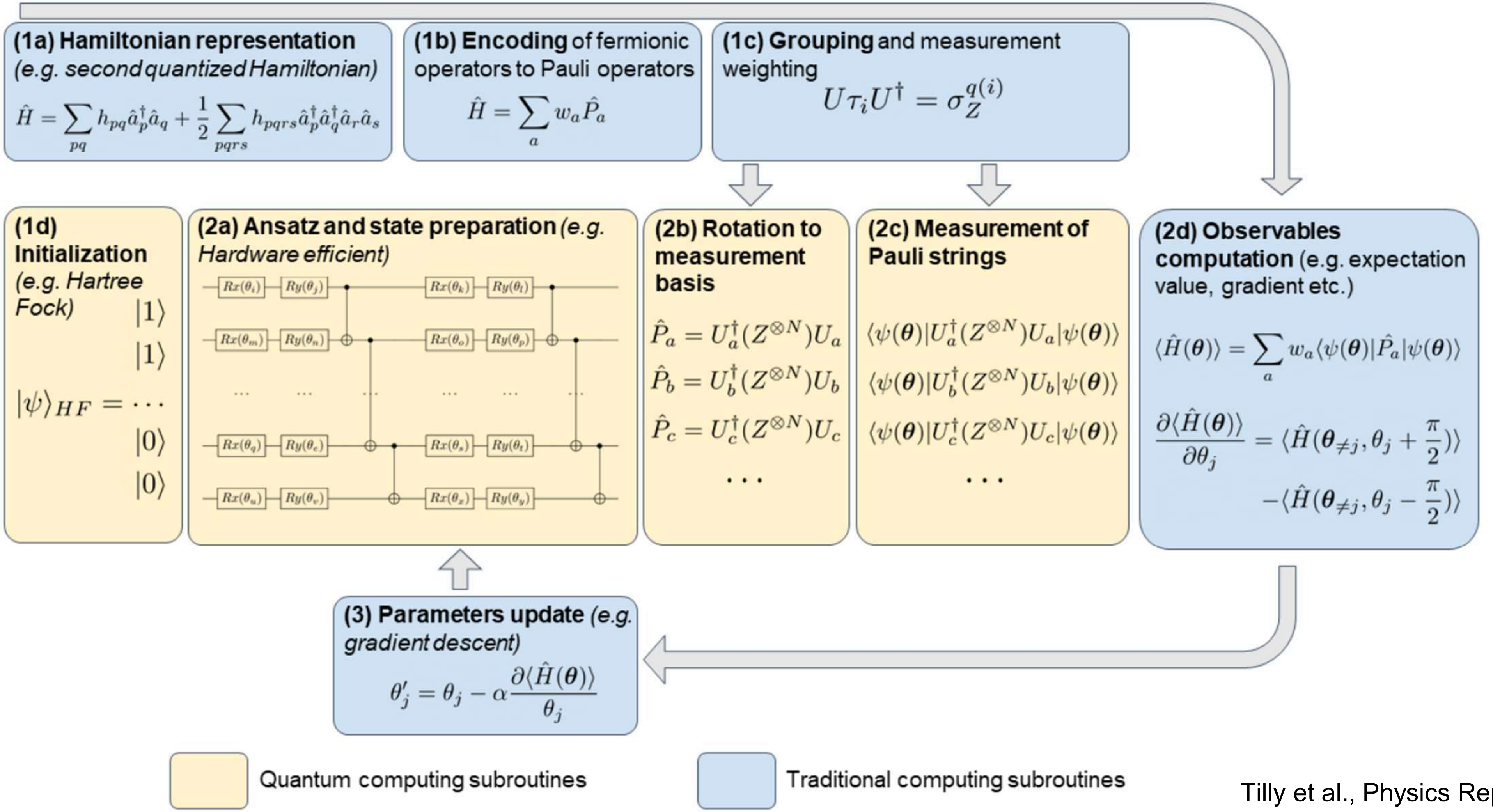
- gradient-based
  - first-order derivative
  - second-order derivative
- gradient-free

e.g. $\theta_j' = \theta_j - \alpha \frac{\partial \langle \hat{H}(\theta) \rangle}{\partial \theta_j}$

A key issue for optimization is the barren plateau problem, where gradients vanish

Many optimizers are borrowed from the deep learning community

Putting it all together → VQE



Tilly et al., Physics Report, 2022

Thank you for your attention

References:

- Tilly et al., Physics Report, 2022
  https://doi.org/10.1016/j.physrep.2022.08.003
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  https://www.mckinsey.com/capabilities/mckinsey-digital/our-insights/quantum-computing-use-cases-are-getting-real-what-you-need-to-know

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